Code No: 156AF

KR20

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(ANAUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech III Year II Semester Regular Examinations, JUNE 2023

INTERNET OF THINGS

(INFORMATION TECHNOLOGY)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks

PART-A (25 Marks)

1. (a)	Define Internet of Things (IoT).	2M
(b)	List the characteristics of IoT.	3M
(c)	List the different types of sensors.	2M
(d)	What are the characteristics of actuators?	3M
(e)	List the data types in Arduino programming.	2M
(f)	What is the interfacing of LCD with Arduino?	3M
(g)	Draw building blocks of an IoT device	2M
(h)	Explain about motion sensors.	3M
(i)	List IoT Application in different domains	2M
(j)	What are smart devices in IoT?	3M

PART- B (50 Marks)

Q.2(a)	What are the communications APIs in IoT? Explain with diagram	5M
(b)	Explain physical design of IOT?	5M
	OR	
Q.3(a)	Explain the logical design of IoT in detail with neat sketch diagram	5M
(b)	Discuss the different communication models in IoT	5M
Q.4(a)	Explain the different types of actuators in IoT	5M
(b)	What are the characteristics of IoT sensors.	5M

6. a) Recall the functions and loops in Arduino Programming. [5M] BTL1
   b) Show the integration of an LED and switch in Arduino Programming. [5M] BTL1

OR

7. Define and Discuss clearly about IoT applications in Arduino Programming. [10M] BTL5

8. Inference Python program for sending Email when control is switched on in Raspberry. [10M] BTL4

OR

9. Distinguish different communication interfaces on the Raspberry Pi board, including I2C, SPI, and UART. [10M] BTL4

10. a) Experiment the Traffic light control system with the help of IoT. [5M] BTL3
    b) Interpret the implementation of data sharing between server and client. [5M] BTL2

OR

11. Design the controller service of the Healthcare Automation in IoT system. [10M] BTL5

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech III Year II Semester Supplementary Examinations, December-2023

INTERNET OF THINGS

(INFORMATION TECHNOLOGY)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A **(25 Marks) BL**

1. a) Tell about Communication protocols in IoT. [2M] BTL1
   b) Define IOT and list out any three characteristics of IoT. [3M] BTL1
   c) Explain the sensor and list any two characteristics. [2M] BTL2
   d) Distinguish IoT and M2M in terms of communication protocols. [3M] BTL4
   e) Develop the interfacing LCD. [2M] BTL3
   f) Discuss about dictionary as a data structure in python. [3M] BTL5
   g) Outline about the IOT device. [2M] BTL2
   h) Identify the importance of Raspberry Pi Interfaces. [3M] BTL3
   i) Examine about the PaaS. [2M] BTL4
   j) Build the Smart Parking IoT System. [3M] BTL5

PART- B **(50 Marks)**

2. a) Explain any two communication models. [5M] BTL2
   b) Demonstrate the Domain specific Applications of IoT. [5M] BTL2

OR

3. a) Model the logical design of the IoT. [5M] BTL3
   b) Construct the SMART CITY With IoT environment by taking any example. [5M] BTL3

4. Classify IoT Communication API's and discuss it in detail. [10M] BTL2

OR

5. a) Tell me about the need of IOT system management. [5M] BTL1
   b) Relate the Ultrasonic sensor and PIR sensor. [5M] BTL1

OR

Q.5(a)	Explain the working principle of pneumatic actuators in detail.	5M
(b)	With a neat diagram, explain how actuators and sensors interact with physical world. Classify actuators based on energy type.	5M
Q.6(a)	Explain the interfacing of LCD with Arduino?	5M
(b)	Give the anatomy of Arduino program.	5M

OR

Q.7(a)	Write a brief note on different types of sensors in Arduino programming	5M
(b)	Write a brief note on arrays and strings in arduino programming	5M
Q.8(a)	Write the python program to control (blink) an LED by using switch.	5M
(b)	Describe the relative strengths and limitations of Building IOT with Raspberry Pi	5M

OR

Q.9(a)	List and explain the basic building blocks of an IOT device	5M
(b)	Explain in detail about RaspberryPi Board	5M
Q.10(a)	Discuss in detail traffic light control system of IoT with neat diagram	5M
(b)	Construct the Design of Smart home with Raspberry Pi and other hardware devices with neat sketch	5M

OR

Q.11(a)	Discuss in detail Smart City application of IoT	5M
(b)	Discuss smart healthcare system using IoT.	5M

Code No:216AD **KR21**

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech III Year II Semester Regular Examinations, July/August-2024

INTERNET OF THINGS

(Information Technology)

Time: 3 hours Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (20 Marks)

1. a) Explain the characteristics of IoT. [2M]L2
   b) Importance of IoT agriculture sector [2M]L2
   c) Identify the important characteristics in IoT sensing [2M]L1
   d) Explain the working of PIR sensor [2M]L2
   e) List any four characters in the functions of Arduino programming [2M]L1
   f) Explain compounding operators in Arduino programming [2M]L2
   g) Explain the purpose of NOOBS software while using Raspberry Pi. [2M]L2
   h) Justify the usage of DHT22 sensor in Raspberry Pi. [2M]L2
   i) Explain the role of IoT in the healthcare sector [2M]L2
   j) Does Raspberry PI have BIOS? Justify your answer [2M]L2

PART- B (5X10=50Marks)

2. Discuss the hierarchy of IoT layers (e.g., perception layer, network layer, application layer) and their respective functions. Provide examples of protocols and standards used at each layer. [10M]L4

OR

3. Describe IoT applications in smart cities. Discuss initiatives such as smart transportation, infrastructure monitoring, and public safetyin an era of IoT. Explain the benefits and challenges of implementing IoT solutions at a city-wide scale. [10M]L4

4. a) Discuss the role of sensors in detecting physical phenomena, converting them into electrical signals, and their importance in various applications. [5M]L4
   b) Explain parameters such as sensitivity, accuracy, resolution, response time, and range. Provide examples of how these characteristics impact sensor performance. [5M]L3

OR

5. a) Discuss the working principle of ultrasonic sensors for distance measurement with examples where ultrasonic sensors are used in IoT applications. [5M]L4
   b) Explain how actuators can be integrated into IoT systems. [5M]L3

6. a) Design Interfacing a Light Emitting Diode (LED) with Raspberry Pi. [5M]L5

   b) Explain use of arrays and strings in Arduino programming. [5M]L3

OR

7. Show in detail about Data Collection and analysis approaches differ in M2M and IoT. [10M]L1

8. a) How SDN can be used for various levels of IoT? [5M]L1

   b) Construct the basic building blocks of an IoT device. [5M]L3

OR

9. a) Explain about the DHT22 sensor with an example. [5M]L2

   b) List and explain the components of Raspberry Pi. [5M]L4

10. Explain in detail about cloud computing services. [10M]L2

OR

11. Explain the Home Automation IoT system with an example. [10M]L4

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Code No: 156AF **KR20**

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech III Year II Semester Supplementary Examinations, July/August 2024

INTERNET OF THINGS

(INFORMATION TECHNOLOGY)

Time: 3 hours Max. Marks: 75

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (25 Marks)

1. a) Develop the physical Design of IoT . [2M]L3
   b) Define IOT and list out any three applications of IoT. [3M]L1
   c) Compare between IoT and M2M with few points. [2M]L2
   d) What are the limitations of SNMP? [3M]L1
   e) Outline about arduino. [2M]L2
   f) What is the use of python programming in IoT. [3M]L3
   g) Relate how SDN can enhance network scalability in an IoT environment. [2M]L2
   h) What are the pins on Raspberry Pi for SPI interface? [3M]L1
   i) What is the sensor cloud? [2M]L4
   j) Discuss about the smart devices. [3M]L5

PART- B (5 X 10 =50 Marks)

2. a) Develop the Functional blocks of IoT. [5M]L3
   b) Label IoT protocols with suitable block diagram. [5M]L1

OR

3. Illustrate about the IoT enabling Technologies in detail. [10M]L2

4. a) Analyze clearly about the PIR Sensor. [5M]L4
   b) Model the Data Handling in IoT in detail. [5M]L3

OR

5. a) Elaborate about the Actuator types. [5M]L5
   b) Explain about the switch and picamera. [5M]L2

6. a) Explain how to control an LED using Arduino by providing a step-by-step example code snippet demonstrating how to turn the LED on and off programmatically. [5M]L5

b) Explain how conditional statements (if, else) and loops (for, while) are implemented in Arduino sketches. [5M]L3

OR

7. a) Discuss how to adjust the LED brightness based on the light intensity sensed by the LDR with an example code to implement this functionality. [5M]L4

b) Explain how to use Serial.begin(), Serial.print(), and Serial.println() functions for serial communication between Arduino and a computer. [5M]L3

8. a) Discuss the Raspberry Pi hardware specifications, operating system options, and the ecosystem of software support. [5M]L4

b) Discuss the circuit setup and demonstrate how to write Python code to toggle the LED state based on the switch input. [5M]L4

OR

9. a) Discuss the principles of Ultrasonic operation and demonstrate how to measure distance using Python programming. [5M]L4

b) Discuss Raspberry Pi processor, memory, GPIO pins, and expansion options. Explain how these features contribute to its flexibility and suitability for diverse IoT applications. [5M]L4

10. a) Discuss communication protocols, data formats, and security considerations involved in transmitting sensor data to a remote server and receiving commands from the server. [5M]L4

b) Discuss the advantages of using Firebase for real-time data synchronization, user authentication, and cloud storage in IoT projects. Provide examples of how Firebase APIs are utilized in Raspberry Pi-based applications. [5M]L4

OR

11. a) Elaborate the implementation of smart devices using Raspberry Pi in IoT ecosystems. [5M]L5

b) Explain how Pi camera captures images, interface Rasperry Pi and Pi Camera to capture and Store images, edit images and capture video. [5M]L3

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Code No:216AD

KR21

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech III Year II Semester Regular Examinations, July/August-2024

INTERNET OF THINGS

(Computer Science and Engineering)

Time: 3 hours   Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (20 Marks)

1. a) Discuss IoT applications in energy management. [2M]L3
b) How does IoT transform the retail industry? [2M]L2
c) Discuss the concept of actuators and their role in converting electrical signals into physical actions. [2M]L3
d) What environmental factors should be considered when deploying sensors? [2M]L2
e) With an example, explain how variables can be used to store and manipulate data in Arduino. [2M]L1
f) Explain the process of interfacing an LCD (Liquid Crystal Display) with Arduino. [2M]L3
g) What are the advantages of using Raspberry Pi in IoT projects? [2M]L1
h) Describe how to integrate push-button switch with an LED on Raspberry Pi. [2M]L3
i) Which parameters are analyzed by IoT devices in the agriculture field? [2M]L4
j) How can Raspberry Pi integrate with cloud platforms (AWS, Google Cloud, etc) for IoT applications? [2M]L3

PART- B (5X10=50 Marks)

2. Discuss the enabling technologies that support IoT deployments. Explain the role of wireless communication protocols (e.g., WiFi, Bluetooth, Zigbee), cloud computing, edge computing, and cyber security measures in IoT implementations. [10M]L4

OR

3. a) Explain how IoT is applied in the logistics sector. [5M]L5
b) Discuss the role of APIs (Application Programming Interfaces) in facilitating interoperability and integration in IoT ecosystems. [5M]L5

4. a) Explain common deviations encountered in sensor readings, such as drift, noise, and offset. [5M]L3
b) Discuss the importance of sensor calibration in minimizing deviations and ensuring accuracy. [5M]L4

OR

5. a) Discuss the operating principle of LDR and explain how LDRs are used as light sensors in automatic lighting systems and other light-sensing applications. [5M]L4
b) Elaborate the concept of actuators and their role in converting electrical signals into physical actions. [5M]L5

6. a) Discuss the use of functions in Arduino sketches to modularize code and improve readability. [5M]L3
   b) Distinguish between arrays and strings in Arduino programming by providing examples demonstrating the creation and usage of arrays and strings. [5M]L4

OR

7. a) Discuss the use of pinMode(), digitalWrite(), digitalRead(), analogWrite(), and analogRead() functions. [5M]L3
   b) Describe the purpose and functionality of the Serial Monitor in Arduino IDE. Explain how to use Serial.begin(), Serial.print(), and Serial.println() functions for serial communication between Arduino and a computer. [5M]L3

8. a) Describe the architecture and key components of the Raspberry Pi board. Discuss its processor, memory, GPIO pins, and expansion options. [5M]L3
   b) Discuss the serial, SPI (Serial Peripheral Interface), and I2C (Inter-Integrated Circuit) interfaces on Raspberry Pi. [5M]L3

OR

9. a) Elaborate the concept of an IoT device, its key characteristics, and how it differs from traditional computing devices. with examples of typical applications. [5M]L4
   b) Discuss the architecture and key components of the Raspberry Pi board. [5M]L3

10. Discuss the key components involved, the algorithms utilized for optimizing traffic flow, and the role of real-time data processing in ensuring efficient traffic management. Provide any examples where Raspberry Pi has been effectively deployed in traffic control systems to illustrate your points. [10M]L4

OR

11. a) Design home automation using Raspberry Pi. [5M]L5
    b) Discuss in depth about the data transfer between client and server in Raspberry Pi based IoT. [5M]L3

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